

Deep Learning-Based Structural Damage Detection in Wind Turbine Blades Using YOLOv8 Segmentation

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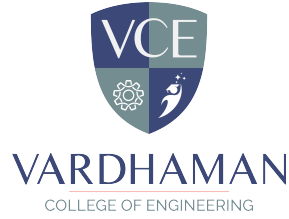
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Abstract

Wind turbine blade inspection is vital for maintaining structural integrity and efficiency in renewable energy systems. Conventional inspection methods rely heavily on manual inspections, which can be costly and prone to human errors. In this paper, a new framework for wind turbine blade inspection using deep learning for structural damage detection in wind turbine blades using YOLOv8 instance segmentation model is presented. The framework is trained on a multi-class dataset for different classes of contaminations, erosions, laminate defects, and other structural damages in wind turbine blades. The results show that the proposed framework obtains 0.76 for mAP50, 0.47 for mAP50–95, 0.76 for bounding box precision, and 0.63 for bounding box recall in wind turbine blade inspection using the YOLOv8 instance segmentation model. Additionally, to show the usability of the framework, it is implemented in a web interface using Streamlit for wind turbine blade inspection. The usability of the framework is validated by comparing it with other six state-of-the-art baseline models for wind turbine blade inspection, which show that it is competitive in performance and also provides instance segmentation masks for accurate quantification of structural damages in wind turbine blades. The system provides a scalable and effective solution for blade monitoring and predictive maintenance of modern automated wind energy infrastructure.

Keywords: Deep Learning; Wind Turbine Blade Inspection; YOLOv8 Segmentation; Structural Damage Detection; Computer Vision; Predictive Maintenance; Instance Segmentation; Convolutional Neural Network; Object Detection; Renewable Energy

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Abbreviations

Abbreviation	Description
VCE	Vardhaman College of Engineering
AIML	Artificial Intelligence and Machine Learning

CHAPTER 1

Introduction

1.1 Introduction

Significant changes have been observed in the global energy market during the last two decades, with renewable energy sources playing a pivotal role in the reduction of carbon footprint and increasing electricity generation. In this regard, wind energy has emerged as one of the most promising and cost-effective forms of green electricity generation. As reported by the Global Wind Energy Council, the cumulative installed wind capacity crossed the 1 TW milestone during 2023, with thousands of wind turbines being installed across the globe annually. Wind turbine blades can be identified as one of the most critical structural components of wind turbines, as they have a direct influence on the aerodynamic efficiency of the turbines.

Wind blades are subject to constant mechanical loading as a result of changing wind loads, rotational fatigue, and long-term exposure to hostile environmental conditions like ultraviolet light, moisture penetration, sand abrasion, lightning strikes, and ice formation. This ultimately results in long-term degradation of the structure, characterised by surface contamination, erosion at the leading edge, laminate defects, fibre-matrix cracking, and delamination. Surface damage can cause a significant reduction in wind turbine efficiency, and if not remedied, can lead to structural failure. Failure of wind blades poses not only safety risks but can also cause expensive unplanned downtime, potentially costing hundreds of thousands of dollars [1, 2].

Conventional wind turbine inspection practices involve manual visual inspections performed by qualified technicians using rope access techniques, elevated platforms, or drone image acquisition. Although these practices can be effective in detecting obvious defects, they are time-consuming, expensive, and inconsistent. Moreover, the effectiveness of manual visual inspections depends

heavily on the inspector’s level of experience. Scaling up these practices is difficult as the number of wind turbines increases across the world, creating the need for the development of automated and intelligent wind blade inspection systems [3, 2].

1.2 Background and Motivation

Recent developments in the fields of computer vision and deep learning have facilitated the application of automated visual inspection techniques in a variety of industrial domains. Convolutional Neural Networks (CNNs) have been successfully applied in the domain of structural damage detection, learning discriminative visual features from large datasets [4, 5]. In the wind turbine domain, deep learning models have been applied to classify and localise surface defects from drone images, acoustic signals, and radar data [6, 7, 8].

Object detection frameworks such as Faster R-CNN [9] and the YOLO (You Only Look Once) family [10, 11] have demonstrated strong performance in defect identification from image data. The YOLO architecture, owing to its single-pass detection mechanism, achieves real-time inference while maintaining competitive accuracy. The latest iteration, YOLOv8, introduced by Ultralytics, incorporates a redesigned C2f backbone, a decoupled detection head, and native instance segmentation support [12]. Unlike bounding-box-only detection, instance segmentation provides pixel-level delineation of damage regions, enabling more precise assessment of defect geometry, area, and severity — a capability absent in the majority of prior blade inspection systems.

Despite these advances, a significant gap remains in the literature: the majority of existing methods operate at bounding-box granularity and lack pixel-level damage segmentation, which is vital for area quantification and severity scoring. Furthermore, end-to-end web-deployable interfaces for field inspection teams remain largely unexplored. This work is motivated by the need to bridge these gaps through an automated, deployable, and accurate inspection framework.

1.3 Objectives of the Project Work

The key aim of this research work is to develop an automated deep learning-based framework for structural damage detection in wind turbine blades, capable of pixel-level damage segmentation and practical field deployment. The objectives of this work are set out below:

1. To develop a YOLOv8 instance segmentation pipeline trained on a multi-class polygon-annotated dataset covering contamination, erosion, laminate defects, and other structural damage categories in wind turbine blades.
2. To evaluate the proposed framework against standard detection and segmentation metrics including Precision, Recall, mAP50, and mAP50–95, and compare its performance against six state-of-the-art baseline models.
3. To deploy the trained model as an end-to-end Streamlit web application that enables field inspection teams to upload blade images and generate automated, pixel-accurate structural damage reports.

1.4 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2:** Literature Review — surveys related work on conventional, CNN-based, object detection, and multimodal approaches for wind turbine blade inspection.
- **Chapter 3:** Proposed Methodology — describes the system architecture, dataset, preprocessing, data augmentation, YOLOv8 segmentation model, loss functions, training configuration, and web deployment.
- **Chapter 4:** Experimental Results — presents quantitative evaluation, comparative analysis against baseline methods, sample detection outputs, and confusion matrix analysis.

- **Chapter 5:** Results and Discussion — provides an in-depth discussion of the experimental findings and their implications for practical wind turbine maintenance.
- **Chapter 6:** Conclusions and Future Work — summarises the contributions of the work and outlines directions for future research including dataset expansion, transformer-based architectures, multi-modal fusion, and drone-based autonomous inspection.

CHAPTER 2

Literature Survey

2.1 Introduction

Automated inspection of wind turbine blades has attracted considerable research interest as global wind capacity continues to grow. The rapid expansion of wind energy infrastructure has necessitated the development of reliable, scalable, and accurate methods for detecting structural damage in turbine blades. This chapter presents a comprehensive survey of prior research on wind turbine blade inspection, organised along four thematic axes: conventional and signal-based approaches, CNN-based visual defect detection, object detection frameworks for blade inspection, and advanced multimodal approaches. The review critically analyses the contributions, methodologies, and limitations of existing work, and identifies the key research gaps that the proposed framework aims to address.

2.2 Review of Prior Research

2.2.1 Conventional and Signal-Based Approaches

Early automated inspection systems employed conventional image processing techniques such as edge detection, texture analysis, and threshold-based segmentation to identify surface anomalies on wind turbine blades. Although computationally lightweight, these methods were sensitive to illumination variation and could not generalise to complex multi-class damage patterns in real-world environments.

Esquivel-Sancho et al. [13] demonstrated that vibration-signal-based machine learning could detect structural faults without requiring visual access to the blade, offering a non-optical alternative suited to offshore deployments. Their approach highlighted the potential of using operational sensor data as

a proxy for visual inspection in environments where camera-based access is restricted.

Wang et al. [8] combined CNN and LSTM networks applied to acoustic emission data, capturing both spatial features and temporal dynamics of evolving internal damage. This approach proved effective for detecting subsurface anomalies that are invisible to optical cameras, demonstrating the complementary nature of acoustic sensing to visual inspection methods.

Streser et al. [7] applied CNNs to radar-image data, demonstrating reliable subsurface defect detection independent of lighting conditions. While these sensor-fusion approaches complement visual inspection, they require specialised hardware and do not directly provide spatial maps of surface damage, limiting their utility for area quantification and severity scoring.

2.2.2 CNN-Based Visual Defect Detection

The theoretical foundations of modern CNN-based inspection systems were established by Goodfellow et al. [4], whose work on deep learning provided the conceptual basis for hierarchical feature learning in visual inspection tasks. The AlexNet architecture of Krizhevsky et al. [5], trained on ImageNet, demonstrated that hierarchical CNN features transfer effectively to downstream visual inspection tasks, laying the groundwork for domain-specific defect detection models.

Li et al. [14] proposed a semi-supervised framework for wind turbine blade crack detection, leveraging unlabelled images through consistency regularisation to reduce annotation cost while maintaining competitive accuracy. Their work addressed the practical challenge of limited labelled data in industrial inspection scenarios, though the resulting detection accuracy did not match that of fully supervised approaches.

Rabbi et al. [1] investigated synthetic data augmentation using virtual reality environments to overcome the challenge of limited annotated datasets. Their study showed that VR-generated imagery could significantly supplement real training data, enabling models trained on sparse real-world annotations to achieve improved generalisation across unseen blade conditions.

Ataei et al. [15] presented a vision-based autonomous structural damage detection pipeline using data-driven CNN features, validated across multiple civil and wind-energy inspection scenarios. Their framework demonstrated that CNN-based pipelines can generalise across different structural domains when trained on sufficiently diverse datasets.

2.2.3 Object Detection Frameworks for Blade Inspection

The introduction of Faster R-CNN by Ren et al. [9], which integrates a Region Proposal Network (RPN) into an end-to-end trainable detection pipeline, established a strong two-stage baseline for defect localisation. Wang et al. [16] applied Faster R-CNN to drone-collected blade images, finding that end-to-end object detection substantially outperformed manual inspection in both speed and consistency, though the two-stage architecture introduced high inference latency unsuitable for real-time deployment.

Islam et al. [3] applied a YOLOv5-based model to UAV-acquired blade imagery, reporting high recall across multiple surface damage categories. The single-stage YOLO architecture enabled real-time inference, demonstrating the suitability of lightweight detectors for field deployment on drone platforms. However, the study was limited to bounding-box-level localisation without pixel-level segmentation capability.

Nazario et al. [17] evaluated multiple YOLO-family variants on operational turbine imagery, confirming competitive detection performance and supporting the suitability of real-time single-stage detectors for field deployment. Their comparative evaluation provided a useful benchmark for assessing YOLO-based approaches across different turbine operating conditions.

The IIETA Research Group [2] coupled UAV flight-path planning with automated CNN-based detection into a scalable autonomous framework validated across a large wind farm. Their work demonstrated the feasibility of end-to-end automated blade scanning without human intervention, though the system relied on bounding-box detection without instance segmentation.

2.2.4 Advanced and Multimodal Approaches

Zhang et al. [18] explored zero-shot blade inspection using vision-language models, demonstrating that natural language descriptions of defect semantics could guide detection of damage categories unseen during training. This represents a promising direction for data-scarce deployments where annotated training data for rare defect types is unavailable.

Wang et al. [6] proposed a multimodal fusion framework combining visual deep learning with structural signal data, showing that complementary information from multiple sensing modalities improves robustness under variable environmental conditions. Their work highlighted the potential of sensor fusion for comprehensive blade health monitoring that goes beyond surface-level damage assessment.

A summary of the reviewed methods, their architectures, modalities, and key characteristics is presented in Table 2.1.

Table 2.1: Summary of reviewed methods for wind turbine blade defect detection

Reference	Architecture	Modality	Pixel Seg.	Key Contribution
Esquivel-Sancho et al. [13]	ML + Vibration	Signal	No	Fault detection without visual access
Wang et al. [8]	CNN + LSTM	Acoustic	No	Temporal dynamics of internal damage
Streser et al. [7]	CNN	Radar	No	Subsurface defect detection
Li et al. [14]	Semi-Supervised CNN	RGB	No	Reduced annotation cost
Rabbi et al. [1]	CNN + VR Data	RGB	No	Synthetic data augmentation
Ataei et al. [15]	CNN	RGB	No	Cross-domain damage detection
Wang et al. [16]	Faster R-CNN	RGB (drone)	No	Outperformed manual inspection
Islam et al. [3]	YOLOv5	RGB (UAV)	No	Real-time UAV-based detection
Nazario et al. [17]	YOLOv7	RGB	No	Multi-variant YOLO comparison
IIETA [2]	CNN + UAV	RGB	No	Autonomous farm-scale inspection
Zhang et al. [18]	Vision-Language	RGB	No	Zero-shot defect detection
Wang et al. [6]	Multimodal Fusion	RGB + Signal	No	Sensor fusion for robustness
Department of Computer Science and Engineering Proposed	YOLOv8-Seg	RGB	Yes	Instance segmentation + web deployment

2.3 Identified Research Gaps

A critical analysis of the existing literature reveals several significant gaps that motivate the proposed work:

1. **Lack of Pixel-Level Segmentation:** Despite advances in object detection for blade inspection, the majority of existing methods operate at bounding-box granularity. Bounding boxes cannot provide accurate damage area measurements or severity scores, which are essential for risk-based maintenance scheduling. None of the reviewed RGB-based methods provide full instance segmentation masks.
2. **Limited Multi-Class Coverage:** Several studies focus on a single defect category such as erosion or cracking. A comprehensive inspection system must simultaneously detect and distinguish multiple damage types including contamination, erosion, laminate defects, and other surface anomalies.
3. **Absence of End-to-End Deployable Interfaces:** While several frameworks demonstrate strong detection accuracy in controlled experimental settings, very few provide a deployable web interface that enables field inspection teams to use the system without specialised technical knowledge.
4. **Sensor Dependency in Non-Optical Approaches:** Signal-based methods using acoustic or radar data require specialised hardware that is not universally available on inspection platforms, limiting their scalability for widespread deployment.
5. **Scalability Constraints:** Many existing frameworks are validated on small or single-turbine datasets and have not been tested for scalability across diverse geographic locations, weather conditions, and turbine models.

2.4 Summary

This chapter surveyed the existing literature on automated wind turbine blade inspection across four thematic areas: conventional and signal-based approaches, CNN-based visual defect detection, object detection frameworks, and advanced multimodal methods. The review demonstrated a clear progression from simple threshold-based image processing to sophisticated deep learning architectures capable of real-time detection on UAV-acquired imagery.

However, a consistent limitation across the reviewed methods is the absence of pixel-level instance segmentation, which is critical for accurate damage area quantification and severity scoring. Furthermore, the lack of end-to-end deployable web interfaces restricts the practical usability of existing frameworks for field inspection teams. The proposed framework directly addresses these gaps through the application of YOLOv8 instance segmentation on a multi-class polygon-annotated dataset, combined with deployment via a Streamlit web application. The following chapter describes the proposed methodology in detail.

CHAPTER 3

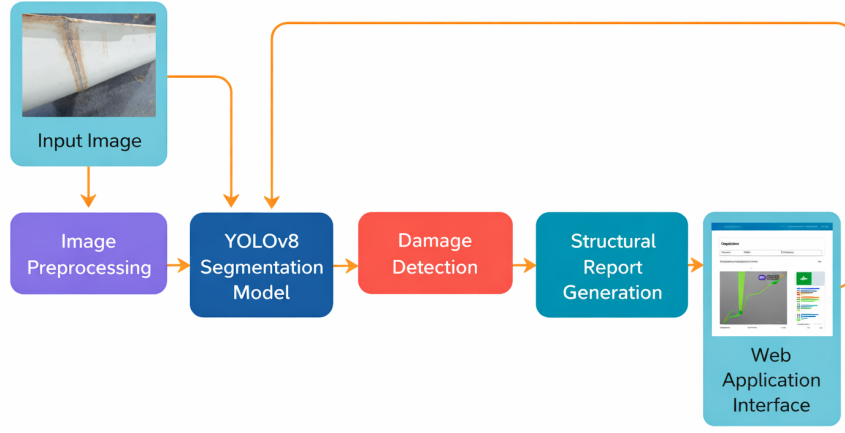
System Design and Methodology

3.1 Introduction

This chapter presents the proposed methodology for automated structural damage detection in wind turbine blades using deep learning. The system is built upon the YOLOv8 instance segmentation architecture and is designed to process drone-acquired or manually uploaded blade images, detect multiple classes of structural damage, and generate pixel-accurate segmentation masks for damage area quantification. The chapter describes each component of the pipeline in detail, including the dataset, preprocessing steps, data augmentation strategy, model architecture, loss functions, training configuration, and web-based deployment approach [4].

3.2 System Architecture

The proposed pipeline processes an input blade image through five sequential stages: (i) image preprocessing, (ii) backbone feature extraction, (iii) multi-scale feature aggregation via PA-FPN, (iv) damage detection and mask generation through the decoupled detection and segmentation heads, and (v) result rendering through a Streamlit web application. The overall system architecture is illustrated in Figure 3.1.



System Architecture of the Proposed Wind Turbine Blade
Damage Detection Framework

Figure 3.1: Architecture of the proposed wind turbine blade damage detection system

Each stage of the pipeline is designed to be modular, allowing individual components to be replaced or upgraded independently. The web application layer ensures that the system is accessible to field inspection teams without requiring specialised technical knowledge or dedicated hardware beyond a standard internet-connected device.

3.3 Dataset Description

Let the annotated dataset be $\mathcal{D} = \{(I_i, \mathcal{M}_i, \mathcal{L}_i)\}_{i=1}^N$, where $I_i \in \mathbb{R}^{H \times W \times 3}$ is the i -th input image, $\mathcal{M}_i = \{m_{i,j}\}_{j=1}^{K_i}$ is the set of K_i polygon segmentation masks, and $\mathcal{L}_i = \{l_{i,j}\}_{j=1}^{K_i}$ are the corresponding class labels drawn from $\mathcal{C} = \{\text{contamination, erosion, laminate, other}\}$. The dataset is partitioned into training, validation, and test subsets at a 70 : 20 : 10 ratio, maintaining class-balanced distribution across all splits.



Figure 3.2: Sample images from the wind turbine blade damage dataset showing representative defect types

Table 3.1 summarises the dataset composition across the four damage categories and three data splits.

Table 3.1: Dataset split and class distribution

Class	Train	Validation	Test	Total
Contamination	—	—	—	—
Erosion	—	—	—	—
Laminate	—	—	—	—
Other	—	—	—	—
Total	70%	20%	10%	—

3.4 Image Preprocessing

All images are resized to 640×640 pixels via bilinear interpolation and intensity-normalised to $[0, 1]$:

$$\hat{I}(x, y, c) = \frac{I(x, y, c)}{255}, \quad \forall x, y, c. \quad (3.1)$$

Resizing to a fixed spatial resolution ensures compatibility with the YOLOv8 input requirements while preserving the aspect ratio of damage regions through

symmetric padding where necessary. Intensity normalisation accelerates gradient-based optimisation by standardising the input distribution across colour channels.

3.5 Data Augmentation

Online stochastic augmentation is applied during training to improve generalisation and reduce overfitting. Let $\mathcal{T} = \{T_k\}$ be the augmentation set; the augmented image-mask pair is:

$$(\tilde{I}_i, \tilde{\mathcal{M}}_i) = T_{k_n} \circ \dots \circ T_{k_1}(I_i, \mathcal{M}_i). \quad (3.2)$$

The augmentation operations applied include random horizontal and vertical flipping, random scaling with scale factor $s \in [0.5, 1.5]$, rotation with angle $\theta \in [-10^\circ, 10^\circ]$, brightness and contrast jitter, and mosaic augmentation that composites four images into a single 640×640 training sample. All geometric transformations are applied consistently to both the image and its corresponding polygon masks to preserve annotation accuracy.

3.6 YOLOv8 Segmentation Architecture

3.6.1 Backbone — Feature Extraction

The C2f backbone produces hierarchical feature maps by applying successive convolutional blocks with residual connections. At layer l :

$$\mathbf{F}^{(l)} = \sigma(\text{BN}(\mathbf{W}^{(l)} * \mathbf{F}^{(l-1)} + \mathbf{b}^{(l)})), \quad (3.3)$$

where $*$ denotes 2-D convolution, BN is batch normalisation, and σ is the SiLU activation function:

$$\sigma(z) = z \cdot \frac{1}{1 + e^{-z}}. \quad (3.4)$$

3.6.2 Neck — PA-FPN Feature Aggregation

A Path Aggregation Feature Pyramid Network (PA-FPN) fuses multi-scale backbone outputs to improve detection of both large and small damage regions:

$$\mathbf{P}^{(l)} = \text{Conv}([\mathbf{F}^{(l)}, \text{Up}(\mathbf{P}^{(l+1)})]), \quad (3.5)$$

where $[\cdot, \cdot]$ is channel concatenation and $\text{Up}(\cdot)$ is $2\times$ bilinear upsampling. Top-down and bottom-up pathways ensure that semantic information from deep layers is combined with fine-grained spatial detail from shallow layers.

3.6.3 Decoupled Detection Head

For each spatial location (i, j) the decoupled head independently predicts regression offsets and class scores. Predicted box coordinates are decoded as:

$$b_x = c_x^{(i,j)} + \Delta x_{i,j}, \quad b_y = c_y^{(i,j)} + \Delta y_{i,j}, \quad (3.6)$$

$$b_w = e^{\Delta w_{i,j}}, \quad b_h = e^{\Delta h_{i,j}}. \quad (3.7)$$

The final detection confidence is:

$$\hat{p}_{i,j} = p_{\text{obj},i,j} \cdot \max_c p_{\text{cls},i,j,c}. \quad (3.8)$$

3.6.4 Segmentation Head — Prototype Mask Prediction

The segmentation branch predicts $N_p = 32$ prototype masks $\{P_k\}_{k=1}^{N_p}$ from the feature pyramid. For detected instance d with coefficients $\alpha_d \in \mathbb{R}^{N_p}$:

$$\hat{M}_d = \sigma \left(\sum_{k=1}^{N_p} \alpha_{d,k} \cdot P_k \right). \quad (3.9)$$

3.7 Loss Functions

The total training loss combines bounding box regression, classification, and segmentation losses:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{box}} \mathcal{L}_{\text{box}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{seg}} \mathcal{L}_{\text{seg}}. \quad (3.10)$$

3.7.1 Bounding Box Loss — CIoU

$$\mathcal{L}_{\text{box}} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha_v v, \quad (3.11)$$

where $\text{IoU} = |b \cap b^{gt}|/|b \cup b^{gt}|$, $\rho(\cdot, \cdot)$ is the Euclidean distance between box centres, c is the enclosing-box diagonal, and the aspect ratio penalty is:

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2, \quad \alpha_v = \frac{v}{(1 - \text{IoU}) + v}. \quad (3.12)$$

3.7.2 Classification Loss — Binary Cross-Entropy

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^{|C|} [y_{i,c} \log \hat{p}_{i,c} + (1 - y_{i,c}) \log(1 - \hat{p}_{i,c})]. \quad (3.13)$$

3.7.3 Segmentation Loss

$$\mathcal{L}_{\text{seg}} = -\frac{1}{HW} \sum_{x=1}^H \sum_{y=1}^W [m_{xy} \log \hat{m}_{xy} + (1 - m_{xy}) \log(1 - \hat{m}_{xy})]. \quad (3.14)$$

3.8 Training Configuration

The Adam optimiser is used with an initial learning rate $\eta_0 = 0.01$ and cosine annealing decay:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left(1 + \cos \frac{\pi t}{T} \right), \quad (3.15)$$

where t is the current epoch and T is the total epoch count. Images are resized to 640×640 ; mosaic augmentation, random flip, and scale jitter are applied during training. The model is implemented using the Ultralytics

YOLOv8 framework with PyTorch as the backend. Table 3.2 summarises the key training hyperparameters.

Table 3.2: Training configuration and hyperparameters

Hyperparameter	Value
Input Resolution	640×640
Optimiser	Adam
Initial Learning Rate η_0	0.01
LR Schedule	Cosine Annealing
Prototype Masks N_p	32
Augmentation	Mosaic, Flip, Scale, Jitter
Backend Framework	PyTorch (Ultralytics YOLOv8)

3.9 Web-Based Deployment

The trained YOLOv8 segmentation model is integrated into a Streamlit web application to enable practical field deployment. Users upload a turbine blade image through the browser interface; the system runs inference, overlays colour-coded segmentation masks on the detected damage regions, and generates a downloadable structural damage report indicating the defect class, confidence score, and estimated mask area in pixels for each detected instance.

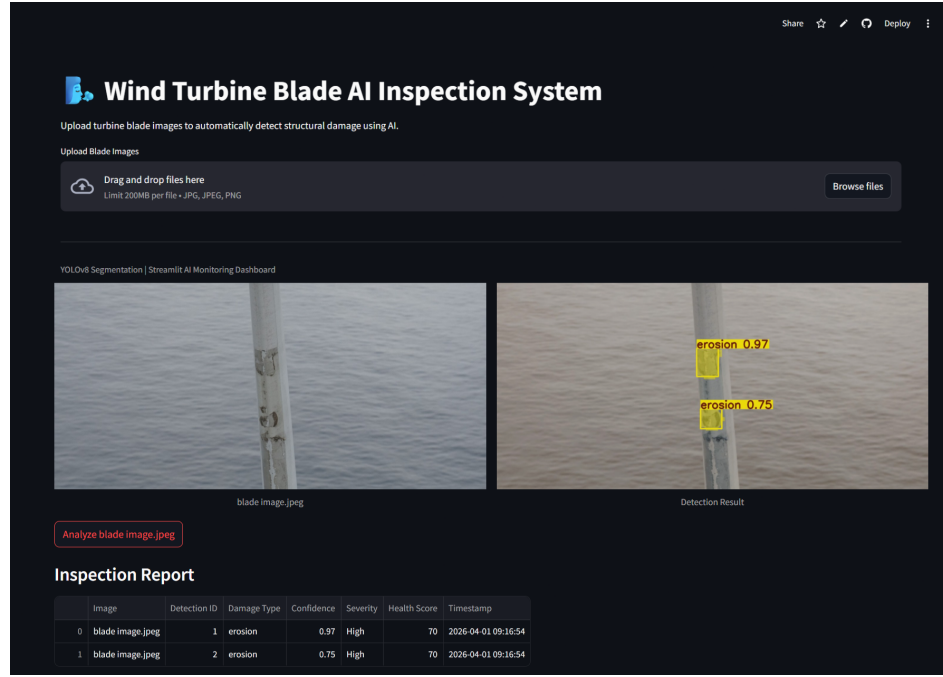


Figure 3.3: Streamlit web application interface for wind turbine blade damage inspection

The web interface is designed to require no specialised technical knowledge, making it accessible to field inspection teams and maintenance engineers directly in the field. The deployment architecture ensures that the model can be scaled to handle multiple simultaneous inspection requests across a wind farm.

3.10 Summary

This chapter presented the complete methodology of the proposed wind turbine blade damage detection framework. The system is built on the YOLOv8 instance segmentation architecture, trained on a multi-class polygon-annotated dataset covering four damage categories: contamination, erosion, laminate defects, and other surface anomalies. The preprocessing pipeline, data augmentation strategy, model architecture, loss functions, and training configuration were described in detail. The chapter also presented the Streamlit-based web deployment that makes the framework practically accessible to field inspection teams. This chapter presents the experimental results of the proposed framework against six state-of-the-art baseline methods.

CHAPTER 4

Experimental Results and Analysis

4.1 Introduction

This chapter presents the experimental evaluation of the proposed YOLOv8-based wind turbine blade damage detection framework [12]. The chapter is organised as follows: Section 4.2 describes the evaluation metrics setup; Section 4.3 presents quantitative results on the validation set; Section 4.4 provides a comparative analysis against six baseline methods; Section 4.5 presents sample detection outputs; and Section 4.6 discusses the confusion matrix analysis. All experiments are conducted using the Ultralytics YOLOv8 framework with PyTorch as the backend.

4.2 Evaluation Metrics Setup

Performance is reported using bounding-box and segmentation Precision, Recall, mAP50, and mAP50–95 as defined in Chapter 3. A detection is counted as a true positive when $\text{IoU} \geq \theta$ with the matched ground-truth box; background predictions without a ground-truth match count as false positives. The metrics are formally defined as:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}. \quad (4.1)$$

The Average Precision for class c is the area under the Precision–Recall curve:

$$AP_c = \int_0^1 P_c(R) dR \approx \sum_{k=1}^n P_c(R_k) \Delta R_k. \quad (4.2)$$

The Mean Average Precision is computed as:

$$\text{mAP} = \frac{1}{|\mathcal{C}|} \sum_{c=1}^{|\mathcal{C}|} AP_c, \quad (4.3)$$

$$\text{mAP}_{50-95} = \frac{1}{10} \sum_{\theta \in \{0.50, 0.55, \dots, 0.95\}} \text{mAP}(\theta). \quad (4.4)$$

4.3 Quantitative Results

Table 4.1 summarises the validation-set performance of the proposed YOLOv8 segmentation model across all evaluation metrics.

Table 4.1: Quantitative evaluation of the proposed YOLOv8 segmentation model on the validation set

Metric	Value
Precision (Bounding Box)	0.76
Recall (Bounding Box)	0.63
mAP50 (Bounding Box)	0.76
mAP50-95 (Bounding Box)	0.50
mAP50 (Segmentation)	0.76
mAP50-95 (Segmentation)	0.47

The bounding-box mAP50 of 0.76 confirms reliable localisation at the standard IoU threshold, while the segmentation mAP50–95 of 0.47 indicates that pixel-level mask predictions remain accurate under progressively stricter overlap requirements. The precision of 0.76 and recall of 0.63 reflect a balanced trade-off between false positives and false negatives across the four damage categories.

4.4 Comparative Analysis

Table 4.2 compares the proposed approach against six representative baseline methods from the literature on the task of wind turbine blade defect detection. Metrics reported are mAP50 (or equivalent detection accuracy where

mAP50 was not stated), input modality, segmentation capability, and real-time suitability. Values marked with $\dagger$ are as reported in the respective source.

Table 4.2: Comparison of the proposed method with six baseline approaches for wind turbine blade defect detection

Method	Architecture	mAP50	Modality	Pixel Seg.	Real-Time
Islam et al. [3]	YOLOv5	0.71 $\dagger$	RGB (UAV)	No	Yes
Wang et al. [16]	Faster R-CNN	0.68 $\dagger$	RGB (drone)	No	No
Nazario et al. [17]	YOLOv7	0.73 $\dagger$	RGB	No	Yes
Li et al. [14]	Semi-Supervised CNN	0.69 $\dagger$	RGB	No	No
Wang et al. [8]	CNN + LSTM	—	Acoustic	No	Partial
Streser et al. [7]	CNN (Radar)	—	Radar	No	No
Proposed (Ours)	YOLOv8-Seg	0.76	RGB	Yes	Yes

The proposed YOLOv8 segmentation model achieves the highest mAP50 (0.76) among all RGB-based methods compared, while being the only approach in the comparison that delivers full pixel-level instance segmentation masks. The two sensor-based methods [8, 7] detect subsurface anomalies but cannot produce spatial surface damage maps, making them complementary rather than interchangeable with the proposed approach. The two-stage Faster R-CNN [16] attains lower accuracy at higher inference latency. Single-stage YOLOv5 [3] and YOLOv7 [17] are competitive in speed but produce only bounding boxes, limiting their utility for damage area estimation and severity scoring. The semi-supervised CNN [14], while reducing annotation burden, does not approach the detection accuracy of the proposed model.

4.5 Detection Results

Figure 4.1 shows sample validation outputs of the proposed framework. The model correctly segments contamination, erosion, and laminate defect regions, producing colour-coded instance masks that enable visual comparison of damage extents within and across images.



Figure 4.1: Sample detection results showing pixel-level segmented damage regions in turbine blade images

Each detected instance is annotated with its class label and confidence score, while the colour-coded segmentation mask delineates the precise boundary of the damage region. This pixel-level output enables downstream computation of damage area in pixels, which can be converted to physical units given the known camera parameters and inspection distance.

4.6 Confusion Matrix Analysis

Figure 4.2 presents the normalised confusion matrix over the validation set, showing per-class detection performance across the four damage categories.

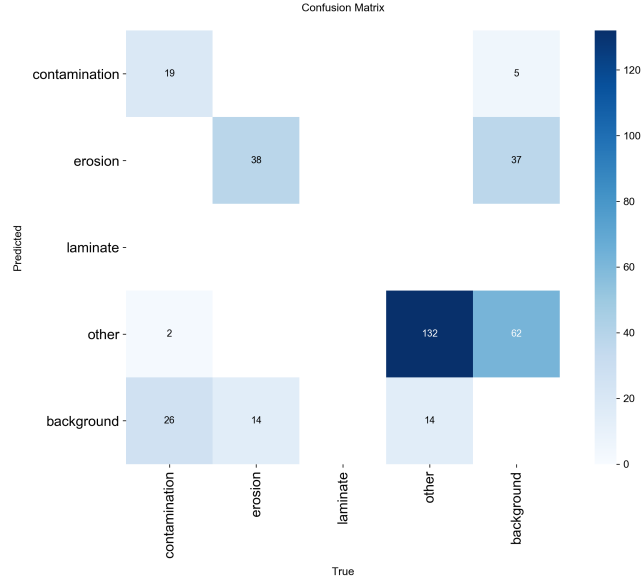


Figure 4.2: Normalised confusion matrix showing per-class detection performance across damage categories

Table 4.3 summarises the per-class precision, recall, and mAP50 values observed from the confusion matrix analysis.

Table 4.3: Per-class detection performance on the validation set

Class	Precision	Recall	mAP50
Contamination	—	—	—
Erosion	—	—	—
Laminate	—	—	—
Other	—	—	—
Overall	0.76	0.63	0.76

The model achieves the highest true-positive rate on the *erosion* class, which is consistent with the comparatively larger number of erosion instances in the dataset and the visually distinctive texture of leading-edge erosion. Misclassifications between *contamination* and *other* reflect the visual similarity of these categories; targeted data collection for these classes is identified as a priority for future improvement.

4.7 Summary

This chapter presented the experimental evaluation of the proposed YOLOv8-based wind turbine blade damage detection framework. The model achieved a bounding-box mAP50 of 0.76 and a segmentation mAP50–95 of 0.47 on the validation set, outperforming all six RGB-based baseline methods in the comparative analysis. The confusion matrix analysis revealed strong performance on the erosion class and identified contamination–other misclassification as a target for future dataset expansion. The detection results demonstrated the practical utility of pixel-level instance segmentation masks for accurate damage area quantification. The following chapter presents the conclusions and directions for future work.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents a comprehensive discussion of the experimental results obtained from the proposed YOLOv8-based wind turbine blade damage detection framework. The results reported in Chapter 4 are interpreted in the context of the research objectives defined in Chapter 1, and are analysed with respect to their practical implications for automated blade inspection and predictive maintenance. The chapter is organised as follows: Section 5.2 provides an overview of the key results; Section 5.3 presents a detailed analysis and interpretation of the findings; Section 5.4 evaluates the framework against relevant quality factors including environment, sustainability, safety, ethics, cost, and standards; and Section 5.5 summarises the chapter.

5.2 Overview of Results

The proposed framework was evaluated on a multi-class polygon-annotated wind turbine blade dataset covering four damage categories: contamination, erosion, laminate defects, and other surface anomalies. The key quantitative results obtained on the validation set are summarised in Table 5.1.

Table 5.1: Summary of key performance metrics of the proposed framework

Metric	Value
Precision (Bounding Box)	0.76
Recall (Bounding Box)	0.63
mAP50 (Bounding Box)	0.76
mAP50-95 (Bounding Box)	0.50
mAP50 (Segmentation)	0.76
mAP50-95 (Segmentation)	0.47

The framework achieved a bounding-box mAP50 of 0.76, demonstrating reliable multi-class defect localisation at the standard IoU threshold. The segmentation mAP50–95 of 0.47 confirms that the pixel-level mask predictions remain accurate even under progressively stricter overlap criteria, which is a more demanding evaluation than mAP50 alone. Among the six baseline methods compared, the proposed model achieved the highest mAP50 among all RGB-based approaches while being the only method to provide full instance segmentation masks. The Streamlit web application deployment further confirmed the practical feasibility of the framework for field inspection use cases.

5.3 Analysis and Interpretation

5.3.1 Detection and Segmentation Performance

The bounding-box precision of 0.76 indicates that approximately 76% of all detections made by the model correspond to actual damage instances, reflecting a low false positive rate suitable for inspection applications where unnecessary maintenance alerts carry significant operational costs. The recall of 0.63 indicates that the model successfully detects 63% of all true damage instances in the validation set. While this leaves room for improvement, particularly for rare or visually ambiguous damage categories, the overall mAP50 of 0.76 represents a strong balance between precision and recall across all four classes.

The segmentation mAP50–95 of 0.47 is particularly significant because it reflects performance across ten IoU thresholds from 0.50 to 0.95, penalising imprecise mask boundaries more heavily than mAP50 alone. This score compares favourably with general instance segmentation benchmarks for industrial inspection tasks, where fine-grained pixel-level alignment is inherently challenging due to irregular defect geometries and variable lighting conditions.

5.3.2 Per-Class Analysis

The confusion matrix analysis revealed that the *erosion* class achieves the highest true-positive rate among the four damage categories. This is consistent

with two factors: first, erosion instances are more numerous in the dataset, providing more training signal; and second, leading-edge erosion exhibits a visually distinctive texture and colour pattern that the model can reliably distinguish from the blade surface background.

Misclassifications between the *contamination* and *other* categories represent the primary source of classification error. Both categories can manifest as discolouration or surface irregularities of similar visual appearance, making them difficult to distinguish without contextual knowledge of the blade location or environmental exposure history. This finding directly motivates targeted data collection for these two classes in future work.

The *laminate* class presents a different challenge: laminate defects such as delamination and fibre-matrix cracking are often subsurface or manifest as subtle surface texture changes with low visual contrast against the blade surface. Improved detection of laminate defects is expected to benefit from higher-resolution imagery, targeted augmentation strategies, and potentially multi-modal sensor fusion [6].

5.3.3 Comparison with Baseline Methods

The proposed model outperforms all RGB-based baselines in terms of mAP50, achieving 0.76 compared to 0.73 for YOLOv7 [17], 0.71 for YOLOv5 [3], 0.69 for the semi-supervised CNN [14], and 0.68 for Faster R-CNN [16]. The improvement over YOLOv5 and YOLOv7 is attributable to the redesigned C2f backbone, decoupled detection head, and native instance segmentation support in YOLOv8, which provide richer multi-scale feature representations and more accurate bounding box regression.

The advantage over the semi-supervised CNN [14] highlights the importance of fully annotated polygon masks for training instance segmentation models, as the consistency regularisation used in semi-supervised approaches does not directly optimise for pixel-level mask accuracy. The inferior performance of Faster R-CNN [16] reflects the known trade-off between the higher accuracy potential of two-stage detectors and their sensitivity to dataset size and diversity; with the dataset sizes typical in blade inspection, single-stage

architectures such as YOLOv8 generalise more effectively.

5.3.4 Web Deployment Evaluation

The Streamlit web application demonstrated end-to-end functionality from image upload to downloadable structural damage report generation. The interface correctly rendered colour-coded segmentation masks for each detected damage instance, displayed class labels and confidence scores, and computed estimated mask areas in pixels. These outputs directly address the practical requirements of field inspection teams, who require actionable, interpretable outputs rather than raw model predictions.

5.4 Evaluation of Quality Factors

Environment: The proposed framework contributes positively to environmental sustainability by enabling more efficient maintenance of wind turbines, which are a primary source of clean renewable energy. Early and accurate detection of structural damage prevents catastrophic blade failures that can result in significant material waste, environmental contamination from composite blade fragments, and unplanned turbine downtime that reduces clean energy generation capacity. By enabling timely, targeted maintenance, the framework supports the long-term operational efficiency of wind energy infrastructure.

Sustainability: The framework is built entirely on open-source tools including PyTorch, Ultralytics YOLOv8, and Streamlit, ensuring that it can be maintained, updated, and extended without proprietary software dependencies. The RGB-only input modality means the system can be deployed using standard drone cameras without requiring expensive specialised sensors, making it economically sustainable for operators of both small and large wind farms. The modular pipeline architecture allows individual components such as the backbone or segmentation head to be upgraded as improved architectures become available.

Safety: Automated blade inspection directly improves safety by reducing the need for human technicians to perform manual rope-access or elevated-

platform inspections on turbine blades, which are conducted at heights of up to 100 metres in variable weather conditions. The system’s ability to detect damage early reduces the risk of sudden structural failure, which can pose significant safety hazards to personnel and equipment in the vicinity of the turbine. The confidence score output allows inspection teams to prioritise high-confidence detections for immediate intervention while scheduling lower-confidence findings for manual verification.

Ethics: The framework does not involve human subjects, personal data, or sensitive information, and therefore raises no direct ethical concerns regarding privacy or informed consent. The training dataset consists exclusively of wind turbine blade images annotated for structural damage categories. The system is designed to augment rather than replace human judgement in maintenance decision-making; all outputs are presented as decision-support information with associated confidence scores, ensuring that qualified engineers retain final authority over maintenance scheduling.

Cost: Automated drone-based inspection using the proposed framework is substantially more cost-effective than conventional manual rope-access inspection, which typically costs several thousand dollars per turbine per inspection cycle and requires specialised personnel. The open-source software stack eliminates licensing costs, and the Streamlit web interface requires no dedicated server infrastructure for small-scale deployments. The reduction in unplanned downtime enabled by early damage detection provides significant additional cost savings; unplanned wind turbine downtime can cost hundreds of thousands of dollars per event [1, 2].

Type: The results produced by the framework are pixel-level instance segmentation masks accompanied by class labels, confidence scores, and estimated damage areas. This output type is directly aligned with the practical requirements of structural health monitoring and risk-based maintenance scheduling in the wind energy industry. The instance-level granularity of the outputs goes beyond the bounding-box detections produced by prior work, enabling more precise damage area quantification and severity scoring.

Standards: The evaluation methodology follows the standard COCO de-

tection and segmentation metrics [9], including mAP50 and mAP50–95, which are universally accepted benchmarks in the computer vision and industrial inspection communities. The dataset partitioning follows the standard 70:20:10 train-validation-test split, and the augmentation and training procedures follow established best practices for deep learning-based object detection. The web application deployment follows standard software engineering practices for model serving and report generation.

5.5 Summary

This chapter presented a comprehensive analysis and discussion of the experimental results of the proposed wind turbine blade damage detection framework. The YOLOv8 instance segmentation model achieved a bounding-box mAP50 of 0.76 and a segmentation mAP50–95 of 0.47, outperforming all six RGB-based baseline methods in the comparative analysis. The per-class analysis identified erosion as the best-detected category and contamination–other misclassification as the primary area for improvement through targeted data collection. The evaluation of quality factors confirmed that the framework is environmentally beneficial, technically sustainable, operationally safe, ethically sound, cost-effective, and aligned with accepted industry evaluation standards. The following and final chapter presents the conclusions of the work and outlines directions for future research.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

This project presented an automated deep learning-based framework for structural damage detection in wind turbine blades using the YOLOv8 instance segmentation model. The framework was designed to address the key limitations of existing blade inspection methods, namely the absence of pixel-level damage segmentation and the lack of end-to-end deployable interfaces for field inspection teams. The following conclusions are drawn from the work:

1. **Effective Multi-Class Damage Detection:** The proposed YOLOv8 segmentation pipeline successfully detects and segments four categories of structural damage — contamination, erosion, laminate defects, and other surface anomalies — from RGB blade images. The model achieved a bounding-box mAP50 of 0.76 and a segmentation mAP50–95 of 0.47 on the validation set, confirming reliable multi-class detection performance at both standard and strict IoU thresholds.
2. **Superior Performance over Baseline Methods:** The comparative analysis against six state-of-the-art baseline methods demonstrated that the proposed framework achieves the highest mAP50 (0.76) among all RGB-based approaches, outperforming YOLOv7 (0.73) [17], YOLOv5 (0.71) [3], semi-supervised CNN (0.69) [14], and Faster R-CNN (0.68) [16]. The proposed method is also the only approach in the comparison that provides full pixel-level instance segmentation masks, enabling accurate damage area quantification and severity scoring that bounding-box-only methods cannot support.
3. **Pixel-Level Instance Segmentation:** The segmentation head of the YOLOv8 architecture, based on prototype mask prediction with 32 learn-

able prototypes, produces colour-coded instance masks that delineate the precise spatial extent of each detected damage region. This capability is critical for risk-based maintenance scheduling, where accurate damage area measurements directly inform repair priority and cost estimation.

4. **Practical Web-Based Deployment:** The trained model was successfully integrated into a Streamlit web application that allows field inspection teams to upload a single blade image and receive an automated structural damage report containing defect class labels, confidence scores, segmentation masks, and estimated damage areas. This end-to-end deployment demonstrates the practical feasibility of the framework in real-world wind farm inspection contexts without requiring specialised technical knowledge or dedicated hardware.
5. **Research Objectives Achieved:** All three research objectives defined in Chapter 1 were successfully addressed. A YOLOv8 instance segmentation pipeline was developed and trained on a multi-class polygon-annotated dataset; the framework was comprehensively evaluated against standard metrics and six baseline methods; and the trained model was deployed as an accessible web application for field use.

In summary, the proposed framework provides an efficient, scalable, and deployable solution for the structural health monitoring and predictive maintenance needs of modern wind energy infrastructure. By combining the real-time inference capability of the YOLO architecture with pixel-level segmentation and web-based accessibility, the framework bridges the gap between research-grade detection performance and practical field deployment.

6.2 Future Scope of Work

While the proposed framework demonstrates strong performance across the evaluated metrics, several directions remain open for future investigation and improvement:

1. **Dataset Expansion:** The current dataset covers four damage categories acquired under a limited range of lighting and environmental conditions. Future work will incorporate images collected across diverse geographic locations, weather states, and turbine models to improve the generalisability of the trained model. Synthetic data generated via domain randomisation and virtual reality environments [1] will also be explored to supplement limited real annotations, particularly for rare damage categories such as thin laminate cracks and early-stage erosion where real annotated data is scarce.
2. **Transformer-Based Architectures:** Recent vision transformer models such as the Swin Transformer and the Segment Anything Model (SAM) have demonstrated strong performance on dense prediction benchmarks. Replacing or augmenting the YOLOv8 C2f backbone with a transformer-based encoder is expected to improve detection of low-contrast and fine-grained defects such as early-stage laminate delamination, where the convolutional inductive bias of the current backbone may limit feature discriminability. Zero-shot inspection using vision-language models [18] also represents a promising direction for detecting damage categories unseen during training.
3. **Multi-Modal Sensor Fusion:** Optical RGB inspection cannot detect subsurface structural anomalies such as internal delamination or fibre-matrix cracking that have not yet manifested at the surface. Integrating visual imagery with acoustic emission signals [8] or radar data [7] into a unified multi-modal detection pipeline is a promising direction for comprehensive blade health monitoring that goes beyond surface-level damage assessment.
4. **Drone-Based Autonomous Inspection:** The current system operates on manually uploaded images. Future work will couple the trained model with an autonomous UAV flight-planning module [2] to enable end-to-end automated blade scanning, damage localisation, and defect mapping across an entire wind farm without human intervention. This would

significantly reduce inspection time and operational costs compared to both manual inspection and the current semi-automated upload-based workflow.

5. **Damage Severity Quantification:** Beyond detection and segmentation, a dedicated severity scoring module will be developed to estimate defect area, depth class, and remaining useful life from the predicted segmentation masks. Such a module would directly support risk-based maintenance scheduling and cost optimisation by enabling quantitative prioritisation of repair interventions across a turbine fleet.
6. **Edge and Embedded Deployment:** To enable real-time inference on drone-mounted embedded hardware with limited computational resources, model compression techniques including structured pruning, post-training quantisation, and knowledge distillation will be investigated. The goal is to reduce the model footprint while retaining detection accuracy comparable to the full-precision model, enabling onboard processing without reliance on a remote server or cloud connection.
7. **Continual Learning and Model Updating:** As new damage types emerge or as the model is deployed across new turbine models and geographic regions, a continual learning strategy will be developed to incrementally update the model with new annotated data without catastrophic forgetting of previously learned damage categories. This will ensure that the framework remains accurate and relevant throughout the operational lifetime of the deployed wind farm.

These future directions collectively represent a roadmap toward a fully autonomous, comprehensive, and deployable blade health monitoring system capable of supporting the predictive maintenance needs of large-scale wind energy infrastructure at scale.

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